| Qn 9 | Answer | Marks |
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| **9(a)** | The magnitude of the electrical force acting between two point charges is proportional to the product of the magnitude of the charges and inversely proportional to the square of the distance between them. | **A1** |
| **(b)(i)** | The electrical force between the electron and proton provides for the centripetal force.<br>$\frac{e^2}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r} \Rightarrow v = \frac{e}{\sqrt{4\pi\varepsilon_0 mr}}$ | **B1**<br>**A1** |

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| **(b)(ii)** | $E_P = -\frac{e}{4\pi\varepsilon_0 r}$<br><br>$E_T = E_P + E_K = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{1}{2}\cdot\frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r}$ | **M1**<br><br>**A1** |
| **(b)(iii)** | Energy $E$ = -13.6 eV = -13.6 x 1.6 x $10^{-19}$ = 2.18 x $10^{-18}$ J<br><br>Radius, $r = -\frac{e^2}{8\pi\varepsilon_0 E} = -\frac{(1.60\times10^{-19})^2}{8\pi(8.85\times10^{-12})(2.18\times10^{-18})} = 5.29\times10^{-11}\text{m} \sim 5 \times10^{-11}\text{ m}$ | **M1**<br><br>**A1** |
| **(c)** | Any similarity from :<br>Both are conservative fields<br>Both have a potential<br>Forces between point charges / point masses obey an inverse square law | **A1** |
| | Any difference from :<br>- gravitational field act on mass but electric fields act on charge<br>- electric fields can be shielded but gravitational field cannot be shielded<br>- force of attraction between masses, but attraction and repulsion between charges | **A1** |
| **(d)(i)1.** | de Broglie wavelength of the Moon, $\lambda_{moon}$ = $\frac{h}{mv} = \frac{6.63\times10^{-34}}{(7\times10^{22})(1\times10^{3})} = 9.47\times10^{-60} = 9\times10^{-60}\text{m}$ | **A1** |
| **(d)(i)2.** | de Broglie wavelength of the electron,<br><br>$\lambda_{electron} = \frac{h}{mv} = \frac{6.63\times10^{-34}}{(9\times10^{-31})(2\times10^{7})} = 3.68\times10^{-11}\text{ m} = 4\times10^{-11}\text{ m}$ | **A1** |
| **(d)(ii)** | <u>De Broglie wavelength associated with the moon (=9 x $10^{-60}$ m) << radius of its orbit</u> and hence the <u>wave nature of moon is insignificant.</u><br><br>OR the de Broglie associated with the electron (=4 x $10^{-11}$ m) comparable to radius of orbit and hence the wave nature is significant. | **A1** |
| **(e)(i)** | As $E = -\frac{dV}{dx}$<br><br>Magnitude of electric field strength between parallel plates *E,*<br><br>$E = \frac{\Delta V}{\Delta x} = \frac{(100\times10^{3})}{(15\times10^{-2})} = 6.67\times10^{5}\text{ N C}^{-1}$<br><br>Electrical force on a charged grain due to the parallel plate<br><br>$F_p = qE = (1.60\times10^{-17})(6.67\times10^{5})$<br><br>$= 1.07\times10^{-11}\text{ N}$ | **M1**<br><br>**A1** |
| **(e)(ii)** | The electrical force between two charged mineral grains which are adjacent to each other. (The distance between the centres of the two grains is equal to the diameter of a single grain.)<br><br>$F_E = \frac{q^2}{4\pi\varepsilon_0 r^2} = \frac{(1.60\times10^{-17})^2}{4\pi(8.85\times10^{-12})(100\times10^{-6})^2}$<br><br>$= 2.30\times10^{-16}\text{ N}$<br><br>From the calculation, we see that <u>the electrical force due to the parallel plate is about $10^5$ times the force between two charged mineral grains</u> and hence, the electrical forces between the grains can be ignored. | **M1**<br><br>**A1**<br><br>**A1** |

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| **(e)(iii)** | Horizontal acceleration, $a_x = \dfrac{F_x}{m} = \dfrac{q\Delta V}{m\Delta x} = \dfrac{(2.00\times10^{-6})(100\times10^{3})}{15.0\times10^{-2}} = 1.33 \text{ m s}^{-2}$ | **A1** |
| **(e)(iv)** | $(\rightarrow)$ $s_x$ = 5.0 cm, $u_x$ = 0 m s$^{-1}$, $a_x$ = 1.33 m s$^{-2}$<br>Using $s_x = u_x t + \dfrac{1}{2}a_x t^2$ ,<br>$\Rightarrow \quad t = \sqrt{\dfrac{2s_x}{a_x}} = \sqrt{\dfrac{2(5.0\times10^{-2})}{(1.33)}}$<br>$= 0.274 \text{ s}$ | **M1**<br>**A1** |
| | **Max Marks** | **20** |